

## IMPLEMENTATION, VERSION CONTROL, AND CODE REVIEW

# Lab09: Multimodal Integration

*Combine CampusConnect RAG, voice, and avatar video into one traceable user journey*

|                                 |                                  |                            |                            |
|---------------------------------|----------------------------------|----------------------------|----------------------------|
| <b>CLASS TIME</b><br>45 minutes | <b>BUILD</b><br>1 vertical slice | <b>TESTS</b><br>2 journeys | <b>REVIEW</b><br>1 peer PR |
|---------------------------------|----------------------------------|----------------------------|----------------------------|

**OUTCOME**

By the end of this lab, your CampusConnect MVP will offer one coherent path: the user can type a supported question, optionally use voice, optionally view the AI presenter, and still reach a safe text or human fallback. Your GitHub branch, commits, pull request, and peer review will show exactly how the integration was made.

## What you are—and are not—building

This is an integration lab, not another generation lab. Reuse the RAG interface already working in CampusConnect, the ElevenLabs Agent created in Week 7, and the HeyGen video created in Week 8. The goal is to connect them without breaking the text experience.

1. Required: CampusConnect RAG remains the source of approved answers and citations.
2. Required: voice and video are optional paths; neither may block the text chatbot.
3. Required: the interface identifies the presenter as AI-generated and exposes a transcript.
4. Not included: n8n, a new knowledge base, a new avatar, voice cloning, or a new feature.

**TOOL-CREDIT RULE**

Do not generate a new ElevenLabs Agent or HeyGen video. In Lovable, make one focused prompt and inspect the proposed changes before accepting them. If Lovable credits are unavailable, make the same small integration in your existing Gradio or repository interface; the deliverables and review criteria remain identical.

## 45-minute clock

| Time      | Work                                                                     |
|-----------|--------------------------------------------------------------------------|
| 0–5 min   | Open the four existing artifacts and create the branch.                  |
| 5–20 min  | Integrate the existing voice and video into the CampusConnect interface. |
| 20–29 min | Run one supported and one unsupported user journey.                      |
| 29–36 min | Commit, push, and open the pull request.                                 |
| 36–43 min | Complete a partner peer review and make one small correction if needed.  |
| 43–45 min | Record the final traceability evidence and submit.                       |

### Step 1 — Open the four existing artifacts (3 minutes)

5. Open [Lovable](#) and select the existing CampusConnect project. Do not start a new project.
6. Open [ElevenLabs Agents](#). Locate the Week 7 CampusConnect Agent; do not create a replacement.
7. Open [HeyGen](#). Locate the Week 8 CampusConnect presenter video; do not regenerate it.
8. Open [GitHub](#) and navigate to your CampusConnect repository.

- ☐ The existing text/RAG interface opens.
- ☐ The existing ElevenLabs Agent opens.
- ☐ The existing HeyGen video plays and its transcript is available.
- ☐ The repository is the same repository used in prior CampusConnect labs.

### Step 2 — Create one focused GitHub branch (2 minutes)

9. In GitHub, use the branch selector. If needed, see [Managing branches](#).
10. Create this exact branch: feature/week-9-multimodal-integration
11. Confirm the new branch name is visible before editing any file.

#### WHY THIS MATTERS

The branch makes the integration traceable and reversible. The Week 7 voice artifact, Week 8 video artifact, and Week 9 code change should not be confused with one another.

### Step 3 — Copy the safe integration material (4 minutes)

12. In ElevenLabs, open the Agent and find its website widget or embed instructions. Use [ElevenLabs widget documentation](#) if needed. Copy only the public widget code or Agent ID intended for browser use.
13. Do not copy an API key, secret, .env value, or private authentication token into Lovable or GitHub.
14. In HeyGen, open the completed Week 8 video. Copy the approved embed code or share URL. Use [HeyGen sharing guidance](#) if needed.
15. Copy the exact Week 8 transcript. Keep the disclosure that the presenter is AI-generated.

**STOP CONDITION**

If the material you are about to paste contains words such as API\_KEY, SECRET, Bearer, token, sk-, or a real student record, stop. Remove the sensitive value and ask the professor before continuing.

### Step 4 — Ask Lovable for one constrained edit (8 minutes)

16. Paste your Week 7 widget code, Week 8 HeyGen source, and Week 8 transcript into the marked locations below.
17. Paste the completed prompt into the existing CampusConnect project in Lovable.

**COPY / PASTE - LOVABLE PROMPT — REPLACE ALL THREE PLACEHOLDERS**

Update my EXISTING CampusConnect interface. Do not rebuild the app and do not replace the current RAG answer logic.

Create one simple user journey with these sections in this order:

1. A heading: "CampusConnect Multimodal Help."
2. The existing text/RAG question-and-answer area remains the primary path.
3. A card labeled "Ask by voice" containing the ElevenLabs Agent widget code I paste below.
4. A card labeled "Watch the orientation" containing the HeyGen embed code or share link I paste below.
5. Directly under the video, show: "AI-generated presenter – not a live staff member."
6. Add buttons or links labeled "Use text instead" and "Read video transcript."
7. Add this fallback message: "If voice or video is unavailable, type your question. Unsupported questions are referred to a human."
8. Do not autoplay audio or video. Keep keyboard navigation and readable contrast.
9. Do not expose API keys, agent secrets, private URLs, student data, or .env values.

ELEVENLABS WIDGET CODE

<PASTE YOUR WEEK 7 WIDGET CODE HERE>

HEYGEN SOURCE

<PASTE YOUR WEEK 8 EMBED CODE OR SHARE URL HERE>

VIDEO TRANSCRIPT

<PASTE YOUR WEEK 8 TRANSCRIPT HERE>

Before changing files, summarize the files you will modify. After the change, list the exact files modified and give four manual checks.

18. Read Lovable's proposed file list. Reject the change if it removes the RAG area, changes the knowledge source, adds autoplay, or inserts a secret.
19. Accept only the focused interface change, then preview the application.

## Step 5 — Inspect the multimodal journey (3 minutes)

20. Start at the top of the CampusConnect page and follow the interface in order.
21. Text/RAG is easy to find and remains the primary answer path.
22. "Ask by voice" is clearly labeled and starts only after the user chooses it.
23. "Watch the orientation" identifies the presenter as AI-generated.
24. The transcript and "Use text instead" path are visible.
25. No audio or video autoplays.
26. If a link or widget fails, keep the text path working and add a plain-language fallback link. A graceful fallback is acceptable; a broken page is not.

## Step 6 — Run two complete user journeys (9 minutes)

Test the journey as a user, not as the person who built it. Record the actual behavior in a short file named docs/week-9/MULTIMODAL\_JOURNEY.md.

| Journey         | Actions and expected evidence                                                                                                                                                                      |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A — Supported   | Ask: “How do I reset my CampusConnect password?” Use text or voice. Confirm a grounded answer, a source/citation, and that the optional video does not interrupt the answer.                       |
| B — Unsupported | Ask: “What are the Thanksgiving dining hall hours?” Confirm that CampusConnect abstains or routes the user to a human. Then mute/skip the video and confirm the same text fallback remains usable. |

### COPY / PASTE - SAVE AS docs/week-9/MULTIMODAL\_JOURNEY.md

```
# Lab09 Multimodal Journey Evidence

Branch: feature/week-9-multimodal-integration

## Journey A — Supported
- Input method used: text / voice
- RAG answer observed:
- Citation or source observed:
- Voice behavior:
- Video remained optional: yes / revise

## Journey B — Unsupported
- Question used: Thanksgiving dining hall hours
- Abstention or human next step observed:
- Text fallback worked with video muted/skipped: yes / revise

## Traceability
- Requirement: one optional multimodal CampusConnect journey
- Files changed:
- Known limitation:
```

## Step 7 — Commit a readable change story (4 minutes)

27. Review the changed-file list. Remove any .env file, credential, generated cache, or unrelated edit.
28. Commit the interface change with: feat: integrate voice and video journey
29. Commit the evidence record with: docs: record multimodal journey checks
30. Push feature/week-9-multimodal-integration to GitHub.

### READABLE COMMITS

Two small commits tell a clearer engineering story than one “updated project” commit: first the implementation, then the evidence used to judge it.

## Step 8 — Open the pull request (3 minutes)

31. Open a pull request from the Week 9 branch into the default branch. If needed, see [Creating a pull request](#).
32. Use this title: Lab09: integrate CampusConnect multimodal journey
33. Paste the body below and replace any statement that is not true for your build.

### COPY / PASTE - PULL-REQUEST BODY

```
## Lab09: Multimodal Integration

### User journey
Text/RAG -> optional voice -> optional avatar orientation -> safe fallback

### Changes
- Integrated the existing ElevenLabs Agent from Week 7
- Integrated the existing HeyGen presenter from Week 8
- Preserved CampusConnect RAG, citations, abstention, and human fallback
- Added disclosure, transcript access, and text-only fallback

### Evidence
- Supported question tested: password reset
- Unsupported question tested: Thanksgiving dining hours
- No secret, .env file, personal data, or autoplay was added

### Reviewer focus
Please verify the complete journey, fallback behavior, disclosure, and secret safety.
```

34. Request review from the partner assigned by the professor. Do not merge yet.

## Step 9 — Complete a six-point peer review (7 minutes)

35. Open your partner's preview and pull request. Review the application before reviewing code.
36. Run or inspect the six checks below. Paste the completed review as one GitHub pull-request comment.

### COPY / PASTE - PEER-REVIEW COMMENT

```
REVIEW RESULT: APPROVE / REQUEST CHANGES

1. RAG answer and citation still work: PASS / REVISE
2. Voice starts only after user action: PASS / REVISE
3. Video disclosure and transcript are visible: PASS / REVISE
4. Text fallback works when voice/video is skipped: PASS / REVISE
5. Unsupported question gives a safe human next step: PASS / REVISE
6. No secret or private data is visible: PASS / REVISE

One specific observation:
One requested change, if any:
```

37. If you request a change, name one observable problem and one expected correction. Avoid comments such as “looks good” or “fix voice.”
38. The author makes at most one small correction, reruns the affected check, and replies with the new commit or observed result.

## Step 10 — Close the traceability loop (2 minutes)

39. In MULTIMODAL\_JOURNEY.md, add the pull-request URL, reviewer name, review decision, and any correction made.
40. Leave the pull request open unless the professor instructs the class to merge.
41. Submit the pull-request URL and one screenshot showing text/RAG, voice, and video/fallback in the same CampusConnect interface.

## Definition of done

- ☐ One interface presents CampusConnect RAG, optional voice, and optional avatar video.
- ☐ The Week 7 Agent and Week 8 video were reused; no new generation was required.
- ☐ The supported journey preserves grounding and citation behavior.
- ☐ The unsupported journey produces abstention or a safe human next step.
- ☐ The presenter disclosure, transcript, text fallback, and no-autoplay behavior are visible.
- ☐ The branch contains focused implementation and evidence commits.
- ☐ A peer completed the six-point GitHub review.
- ☐ No key, secret, .env file, private record, or unrelated change was committed.

## Submission and grading

| Evidence                                   | Points |
|--------------------------------------------|--------|
| Working multimodal CampusConnect journey   | 35     |
| Supported and unsupported journey evidence | 25     |
| Secret-safe branch and readable commits    | 15     |
| Pull request and substantive peer review   | 20     |

| Evidence            | Points |
|---------------------|--------|
| Complete submission | 5      |

**NEXT WEEK**

Week 10 treats this integrated MVP as a release candidate. You will separate deterministic software checks from probabilistic AI evaluation, add a lightweight CI gate, repair severe defects, and decide whether the product is ready to freeze.